Appendix C. Core Data Structures

You can't build a house from the roof. Likewise, you can't build knowledge on advanced data structures if you don't master the basics first.

In this appendix, we provide an overview of core data structures, and present some of the most widely used algorithms.

It starts with are a refresher of the most basic data structures: arrays, linked lists, trees, and hash tables. We do expect readers to be already familiar with them: after all, these structures are the building blocks of the most advanced ones. But we'll provide a quick overview, just for your convenience.

In the last section, we briefly compare these data structures: for each of them, we will look at key features (like if they support ordering or not, if they are static or dynamic) and sum them up in a table: this will help us decide, for each problem we might run into, which one of them would be more appropriate to support us in solving the problem.

C.1 Core Data Structures

Data structures are one of the foundations of programming, and have been progressively introduced since the very dawn of computer science.

In this section, we are going to explore the most basic ways to organize data items in memory, such that those items can later be retrieved according to specific criteria. The nature of those criteria, together with the way storage is used and the performance of the basic operations (adding, removing and searching elements) are what determines the characteristics of a data structure.

These data structures are the building blocks for the implementations of countless advanced ADTs.

C.1.1 Array